

Where Atlas & Vine Loses Its Customers

A diagnostic of fulfillment, satisfaction, and retention across 99,441 orders, identifying which problems are fixable and which are structural.

PREPARED BY AKSHARA · JUNE 2026 · DATASET: 99,441 ORDERS · 96,096 CUSTOMERS · 11,997 REVIEWS ANALYZED

Atlas & Vine entered this analysis with three symptoms: rising returns, falling repeat purchases, and a growing support backlog. The question was where customers drop off, what frustrates them, and what is worth fixing. The honest answer splits cleanly in two. **Delivery is a real, measurable lever the business can pull. Retention is a structural problem that no per-customer model can target away.** Everything below is sized so the team knows where to spend and where not to.

01 The five findings

1 · Late delivery is the single clearest satisfaction lever

On-time orders average a **4.21** review; late orders crash to **2.27** — a **1.94-star** drop. **62.4%** of late orders score 1–2 stars, versus 11.3% when on time. This is the cleanest, hardest finding in the project and it covers the full order population, not a sample.

2 · Satisfaction holds for three weeks, then falls off a cliff

Reviews stay healthy up to ~21 days (0–7d: 4.41, 8–14d: 4.29, 15–21d: 4.10) then collapse to **3.01** at 22 days or more. The damage isn't gradual — there's a clear breakpoint past three weeks.

3 · Roughly 3% of orders never reach the customer

2,965 orders (2.98%) drop out of the fulfillment funnel. The largest single leak is the **approved → handed to carrier** step (1,623 orders stuck in canceled, unavailable, or processing). This is silent churn — customers who paid and got nothing.

4 · Retention is structurally low — this is the headline problem

Only **3.12%** of customers (2,997 of 96,096) ever place a second order. **96.9%** are one-and-done. This is larger than the delivery problem and it reshapes everything downstream — it collapses the frequency dimension of segmentation and rules out a conventional churn model.

5 · First-order experience does not predict whether a customer returns

A leakage-free repeat-purchase model (XGBoost, features built from the first order only) scored **ROC-AUC 0.59** — barely above random. Delivery speed, basket size, and review score none meaningfully move the odds of a repeat. **This is a finding, not a failed model:** the low score is trustworthy precisely because the design was leakage-free, and it tells us the 3% repeat rate is structural, not something individual targeting can fix.

The two findings that matter most point in opposite directions: fix delivery for everyone, and stop trying to predict retention for anyone.

02 What customers actually complain about

Of a complaint corpus of 11,997 reviews, the theme split is **delivery 63%** and **quality 25%** — together ~87% of complaints. Among late orders specifically, delivery rises to 82% of complaints and drowns out everything else. Clustering the negative reviews surfaces one overwhelming message: "never received it" / "still haven't gotten my order" is by far the dominant cluster, ahead of "arrived defective" and post-office pickup friction.

Read this carefully: this corpus was deliberately over-sampled toward 1–2 star reviews (86% of it), so these percentages describe what complainers complain about — not the share of all customers who are unhappy. Population unhappiness comes from the review-score distribution, not from this set.

03 Who the customers are

Behavioral segmentation (K-means on spend, recency, and delivery experience; frequency excluded because 97% of customers have exactly one order) produces four actionable groups:

Segment	Customers	% base	Avg spend	Avg delivery	Late rate	Avg review	Repeat
High-value	27,065	28.2%	R\$335	12.3 d	0%	4.12	7.0%
Low-value, recent	34,611	36.0%	R\$73	8.9 d	0%	4.30	1.1%
Dormant	28,097	29.2%	R\$117	10.8 d	0%	4.19	—
Late & unhappy	6,323	6.6%	R\$179	33.5 d	99%	2.27	—

- **High-value** is 28% of customers but ~57% of observed revenue (R\$9.07M of R\$16.0M) and the only segment that repeats above the 3% base rate. Retention upside is concentrated here.
- **Late & unhappy** is mid-value (spends more than the low-value majority) and being destroyed by 33-day deliveries — a fixable loss, not low-value churn.

04 Recommendations

1 • Fix delivery as a system, not customer by customer

Target the two specific failure modes the data names: the **approved → carrier handoff leak** (1,623 lost orders) and the **22-day-plus tail** where satisfaction collapses. This is the one lever with a proven satisfaction effect (1.94 stars). Owner: Operations.

Assumption: the late-delivery → low-review relationship is causal, which the population-level gap (4.21 vs 2.27 across 98K orders) strongly supports but a controlled test would confirm.

2 • Protect and deepen the high-value segment

27,065 customers, 57% of revenue, the only group repeating above base. Concentrate any retention or loyalty spend here rather than spreading it thin across the one-time majority. Owner: Marketing / CRM.

Assumption: past observed value is a reasonable proxy for future value; note this is descriptive CLV, not a predictive lifetime model (the repeat base is too thin to fit one).

3 • Do not build a per-customer churn or at-risk targeting program

The repeat model's AUC of 0.59 means any ranked "at-risk" list would be ~96% wrong. Spending on individual retention scoring is unsupported by the data. Redirect that budget to the systemic delivery fixes in Recommendation 1, which move the metric that demonstrably matters.

This is the contrarian call and the most defensible one: the leakage-free design is exactly why the weak result is trustworthy.

05 Sizing the prize (honestly)

This analysis intentionally does **not** publish a single "X% churn reduction" figure, because there is no intervention or causal-test data to support one and inventing it would be dishonest. What can be stated is the size of the addressable problem:

- **~6,300 customers** sit in the Late & unhappy segment and ~6,350 orders ran late — the directly addressable population for the delivery fix.
- **R\$9.07M** of observed revenue concentrates in the 27K high-value customers worth protecting first.
- **~2,965 orders** never delivered represent pure, recoverable fulfillment loss.

The right next step before claiming an impact number is a small controlled delivery-improvement pilot on a region or carrier, measuring the review-score lift directly. That converts a correlation into a defensible projection.

Method notes & honest limitations

- **Data & currency:** figures are drawn from the Olist Brazilian e-commerce dataset; all monetary values are in Brazilian reais (R\$). "Atlas & Vine" is a narrative framing for the analysis, not a separate data source.
- **Topic modeling** is TF-IDF + K-means clustering, not BERTopic (the transformer backend was offline during the run). Clusters are coherent but coarser than a neural approach.
- **Emotion tags** are rule-based keyword matching, not a fine-tuned model — directional only. The high "frustration" share is partly a default-bucket artifact and should not be quoted as a precise figure.
- **Sentiment / theme classifiers** are TF-IDF + logistic regression (84% sentiment accuracy, delivery-theme F1 0.76). Minority themes are data-limited and directional.
- **CLV is descriptive** (observed historical value), not a predictive lifetime model — the 3% repeat base can't support one.
- **The complaint corpus is over-sampled toward negatives by design.** It answers "what do people complain about," never "what fraction of customers are unhappy."
- **The companion dashboard is interactive, not live** — it recomputes in-memory from committed analytical datasets, not from a production database.